
IAL Physics Unit 4

Rapid Revision

WPH14 — Further Mechanics, Fields and Particles

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* study note
(22 past papers and mark schemes, June 2019 - January 2026)

90 marks · 1 h 45 min · Section A: 10 multiple choice · Section B: 80 marks
1-2 asterisked (*) questions, always 6 marks

≈1.15 min per mark. Section A in ≤12 minutes, leaving ~90 min for 80 marks.

TRIAGE — WHERE THE MARKS ARE

Topic	Marks/paper	Appears
Momentum, impulse, 2D collisions	12–18	Every paper. These six carry the bulk of Section B. Secure them first.
Capacitors (incl. exponential charge/discharge)	9–18	
Circular motion	9–17	
Electric fields, potential, Coulomb	7–16	
Electromagnetic induction	6–17	
Particle physics, conservation laws	5–16	
Magnetic force on wires / charges	3–12	Every paper
Alpha scattering	3–13	~90%
Accelerators (linac / cyclotron)	4–13	~85%
Relativistic effects (muons)	2–7	~70%
de Broglie / electron diffraction	3–4	~60%

THE THREE THINGS THAT SEPARATE A* FROM A

- 1. Command-word discipline.** Most Section B marks are lost by doing a correct calculation and then **failing to write the one sentence of comparison and conclusion** the scheme demands.
- 2. Mark-scheme vocabulary.** "change in flux linkage" · "cuts lines of magnetic flux" · "opposes the change that caused it" · "no resultant external force" · "concentration of charge". Paraphrases often fail.
- 3. 2D momentum and exponential capacitor work done fast and cleanly** — the two biggest blocks of marks, and where time is lost.

THE SINGLE HIGHEST-YIELD HABIT

Every **Deduce / Discuss / Assess / Evaluate** answer must end with:

"[my value] is [greater than / less than / equal to] [the target], so [the claim is correct / the collision is inelastic / the design does meet the requirement]."

This final mark is awarded on **error carried forward** — it is free even if the arithmetic went wrong.

SHEET 1 — MOMENTUM & IMPULSE

$p = mv$ (vector, kg m s⁻¹ = N s) · **impulse** = $F\Delta t = \Delta p$ · $F = \Delta p/\Delta t$ · $E_k = p^2/2m$ · **area under an F-t graph = impulse = Δp .**

THE REBOUND TRAP — APPEARS CONSTANTLY

Ball mass m hits a wall at u , rebounds at v in the **opposite** direction:

$\Delta p = m(-v) - m(u) \rightarrow$ **magnitude = $m(u + v)$**

Same speed $\rightarrow |\Delta p| = 2mv$, so $F = 2mv/\Delta t$. Half speed $\rightarrow |\Delta p| = 3mv/2$, so $F = 3mv/(2\Delta t)$.

Force on the wall is **equal and opposite** to the force on the ball (Newton's third law). **State your sign convention.**

CONSERVATION OF MOMENTUM — THE EXACT WORDING, BOTH HALVES NEEDED

"The **total momentum of a system remains constant, provided no resultant external force acts.**"

The assumption question: no resultant external force acts, **because** an external force would give an extra impulse, so the change in momentum would be different.

Why an air track: friction is negligible, **so there is no resultant external force** / the system is closed.

Rate-of-change-of-mass (jets, propellers, helicopters): $F = (\Delta m/\Delta t) \times v$. Worked pattern: $A = \pi r^2 \rightarrow s = vt \rightarrow V = As \rightarrow m = \rho V \rightarrow p = mv \rightarrow F = p/t$.

Helicopter explanation (4 marks): blades give the air **downward momentum** · force on air = **rate of change of momentum** of the air · by **Newton's third law** the air exerts an equal **upward** force on the helicopter · this equals the weight, so the resultant force is zero and it stays at constant height.

SHEET 2 — 2D COLLISIONS (every paper, 8-12 marks — the highest-value skill)

FORMAT 1 — RESOLVE INTO COMPONENTS (usually fastest)

Along the original direction: $p_1 = p_2 \cos \theta_2 + p_3 \cos \theta_3$

Perpendicular to it: $0 = p_2 \sin \theta_2 - p_3 \sin \theta_3$

Divide perpendicular by parallel — this eliminates the unknown speed and gives θ directly:

$\tan \theta = (m_A v_A \sin \alpha) \div (m_A u_A - m_A v_A \cos \alpha)$

A* habit: always write the two equations **explicitly labelled** "along original direction" and "perpendicular to original direction" — that alone often earns 2 marks even if the arithmetic fails. Carry unrounded values; round at the end.

Format 2 — scaled vector diagram (when the question says to use one): **state a scale explicitly** ("1 cm = 0.10 N s") · draw momenta **head to tail with arrows** · label every vector with quantity and magnitude · the initial momentum is the **closing side** · measure and convert · **finish with a comparison sentence:** "the measured resultant is __, which is (not) equal to the initial momentum of __, so momentum is (not) conserved."

Why it works: momentum is a vector, so if it is conserved the vector sum of the final momenta equals the initial momentum — the vectors form a **closed triangle**.

Format 3 — decay from rest: momentum vectors form a **closed polygon** (total zero). **2 products** $\rightarrow$ exactly opposite directions, equal magnitudes. **3 products** $\rightarrow$ closed triangle; solve for $p \cos \theta$ and $p \sin \theta$, then $p = \sqrt{(\dots)}$ and $\theta = \tan^{-1}(\dots)$. **Give magnitude AND direction** — the direction mark is separate.

ELASTIC vs INELASTIC — THE DEDUCTION, WORTH 3-5 MARKS EVERY PAPER

Elastic: kinetic energy is conserved. Inelastic: E_k is not conserved — transferred to internal energy and sound. **Momentum is conserved in both.**

1. Total E_k before · **2.** total E_k after · **3. explicit numerical comparison + a sentence:** " E_k before = 127 J, E_k after = 112 J. E_k is not conserved, so the collision is inelastic."

Shortcut: equal masses, one initially stationary, elastic $\rightarrow$ the two move off at **90°** to each other. A measured 91° means **slightly inelastic**.

If told "the collision is elastic", use **conservation of E_k as an extra equation**.

Glider results (A hits stationary B, elastic): $m_B < m_A \rightarrow$ A continues slower, B faster · $m_B = m_A \rightarrow$ **A stops dead**, B takes A's velocity · $m_B > m_A \rightarrow$ A rebounds, B moves forward slowly.

SHEET 3 — CIRCULAR MOTION

Quantity	Equation	Note
Angular velocity	$\omega = \theta/t = 2\pi/T = 2\pi f$	rad s ⁻¹ . $\omega = 2\pi f$ is NOT on the formula sheet
Linear speed	$v = \omega r$	Same ω everywhere on a rigid body, so larger $r \rightarrow$ faster
Centripetal acceleration	$a = v^2/r = r\omega^2 = v\omega$	Directed towards the centre
Centripetal force	$F = mv^2/r = mr\omega^2$	The resultant of the real forces

Conversions (guaranteed MCQ): N rev per minute $\rightarrow \omega = N \times 2\pi/60$ · N rev in t seconds $\rightarrow \omega = N \times 2\pi/t$.

DERIVE $a = v^2/r$ — 5 MARKS, GUARANTEED, MEMORISE ALL FIVE STEPS

1. Vector diagram: v_A and v_B (equal magnitude, tangential) drawn head to tail, third side labelled Δv — the diagram is $\mathbf{v}_B + (-\mathbf{v}_A) = \Delta \mathbf{v}$.
 2. **Small-angle statement:** for small θ , $\Delta v/v \approx \theta \approx \sin\theta$ (arc $\approx$ chord).
 3. $\theta = \omega t$ and $v = r\omega$, so $\theta = \mathbf{v}t/r$. 4. $a = \Delta v/t$.
 5. $\Delta v/v = \mathbf{v}t/r \rightarrow \Delta v = v^2t/r \rightarrow \mathbf{a} = \Delta \mathbf{v}/t = \mathbf{v}^2/r$
- Students routinely lose the **small-angle** mark and the **vector-diagram** mark.

FREE-BODY DIAGRAMS — THE SINGLE MOST COMMON ERROR

Do NOT draw "centripetal force" as a separate arrow. It is the **resultant** of the real forces, not an extra one. **Never draw an outward force.**

Arrows start on the body, roughly correct relative lengths, **labelled with names** — weight, tension, normal contact force, friction, lift, drag — not symbols only.

Geometry	Equations and the explanation that scores
Conical pendulum / banked aeroplane	Vertical: $\mathbf{T} \cos\theta = \mathbf{mg}$ · Horizontal: $\mathbf{T} \sin\theta = \mathbf{mv}^2/r \rightarrow \tan\theta = \mathbf{v}^2/(rg) = \mathbf{r}\omega^2/g$. With $r = l \sin\theta$: $\omega = \sqrt{(g/(l \cos\theta))}$. For a banked track with diameter d , $r = d/2$ so $\tan\theta = 2v^2/(gd)$.
Banking — how it allows horizontal circular flight (4 marks)	The vertical component of lift equals the weight , so the vertical resultant is zero and it does not accelerate vertically · the horizontal component provides the centripetal force , acting at 90° to the motion · so the aeroplane follows a circular path. Level the wings at the same lift $\rightarrow$ lift is entirely vertical and greater than the weight $\rightarrow$ resultant upward force $\rightarrow$ it gains height.
Vertical circles	Top: $mv^2/r = mg + N \rightarrow N = mv^2/r - mg$ · Bottom: $N = mv^2/r + mg$. Speed constant $\rightarrow$ magnitude of centripetal force constant ; contact force is maximum at the bottom, minimum at the top . Minimum speed: limiting case $N = 0$, so $mg = mv^2/r \rightarrow v_{\min} = \sqrt{(gr)}$.
Friction providing F_c	$F_{\text{friction}} \geq mv^2/r = mr\omega^2$. Note m cancels — a full and an empty glass are equally likely to slide.

THE CENTRIFUGAL-FORCE "ASSESS THE CLAIM" QUESTION — GUARANTEED, 4 MARKS

By **Newton's first law**, without a force the passenger would continue in a straight line; a force is needed to **change direction** · the force on the passenger acts **towards the centre** · that inward force is exerted by the **car (seat/door) on the passenger** · **there is no outward force on the passenger, so the claim is incorrect** — the sensation is the car pushing inwards.

SHEET 4 — ELECTRIC FIELDS

$E = F/Q$ — "the force per unit positive charge" (vector; $N\ C^{-1}$ or $V\ m^{-1}$) · radial $E = kQ/r^2$ · uniform $E = V/d$ · $F = kQ_1Q_2/r^2$, $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9$.

$V = kQ/r$ — scalar, so potentials from several charges **add algebraically with sign**. $V \propto 1/r$, $E \propto 1/r^2$ — **never confuse these**. $W = Q\Delta V$ · $E_p = kQ_1Q_2/r$ · area under an E-r graph from r to infinity = **the potential at r** · isolated sphere $C = 4\pi\epsilon_0 r$.

Drawing	The exact marking points
Point charge (3 marks)	4-6 straight radial lines, evenly spaced, symmetric · arrows out for +, in for - · lines meet the surface radially and do not stop short or cross
Parallel plates	Straight, parallel, equally spaced · perpendicular to the plates · arrows from + to - · Edge effects allowed, never required
Equipotentials	Always perpendicular to field lines . Radial: concentric circles that get further apart as r increases ($V \propto 1/r$) — drawing them equally spaced loses a mark . Uniform: straight lines parallel to the plates, equally spaced. No work is done moving along an equipotential .

DISTANCE OF CLOSEST APPROACH — NEAR-GUARANTEED

1. Convert E_k : $MeV \times 1.6 \times 10^{-13}$, or $eV \times 1.6 \times 10^{-19}$.

2. **All initial kinetic energy converts to electric potential energy** at closest approach.

3. $E_k = Q_\alpha Q_{\text{nucleus}}/(4\pi\epsilon_0 r)$, with $Q_\alpha = 2e$, $Q_{\text{nucleus}} = Ze \rightarrow r = 2Ze^2/(4\pi\epsilon_0 E_k)$.

4. Then $F = kQ_1Q_2/r^2$, and $a = F/m_\alpha$ if asked.

Assumptions to state: the nucleus behaves as a **point charge**; the nucleus **remains stationary**; the alpha is a point charge.

The twist: an alpha deflected through a **smaller angle** does not get as close — it **retains kinetic energy** at closest approach, so the force and acceleration are smaller.

Electron gun: $eV = \frac{1}{2}mv^2 \rightarrow v = \sqrt{(2eV/m)}$. Do this in your sleep. Must be in a **vacuum** so electrons are not absorbed or scattered by air molecules. **Thermionic emission:** the filament is heated by a current; electrons gain kinetic energy; some gain enough to **escape from the surface**.

Parabolic deflection: $E = V/d \rightarrow F = EQ = VQ/d \rightarrow a = F/m$ · $t = L/v_{\text{horiz}}$ · $s_1 = \frac{1}{2}at^2$ **between** the plates · $v_y = at$ · $s_2 = v_y t/2$ to the screen · **total = $s_1 + s_2$** — forgetting s_1 is the standard error.

Why parabolic: the field is **uniform so the force is constant** · the force is **perpendicular to the initial velocity**, so the horizontal velocity is constant · constant vertical acceleration + constant horizontal velocity $\rightarrow$ parabola.

Millikan: stationary drop $QV/d = mg \rightarrow Q = mgd/V$; then divide by 1.6×10^{-19} and comment on whole-number multiples. Terminal velocity: $6\pi\eta r v = \rho(4/3)\pi r^3 g \rightarrow r = \sqrt{(9\eta v/(2\rho g))}$.

Velocity selector: $QE = BQv \rightarrow v = E/B = V/(Bd)$.

SHEET 5 — CAPACITANCE

$C = Q/V$ (farad = $C\ V^{-1}$; base units $kg^{-1}\ m^{-2}\ A^2\ s^4$) · $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}Q^2/C$ · on a **Q-V graph the gradient is C and the area is the energy** (on a V-Q graph C is the **inverse** of the gradient).

Why $\frac{1}{2}QV$ not QV : the battery does work QV , but the capacitor stores only $\frac{1}{2}QV$ because **the p.d. rises from 0 to V as it charges** — half the energy is dissipated in the circuit resistance.

COMBINATIONS — NOT ON THE FORMULA SHEET

Parallel: $C = C_1 + C_2$ (same V, charges add) · **Series:** $1/C = 1/C_1 + 1/C_2$ (same Q, p.d.s add).

Two identical in series: total C halves, so **total charge halves — and each capacitor carries that same charge**, because charge on series capacitors is equal.

Charge sharing: $Q_{total} = C_1V_0$, then $V = Q_{total}/(C_1 + C_2)$.

Discharging	Charging
$Q = Q_0e^{-(t/RC)}$ $V = V_0e^{-(t/RC)}$ $I = I_0e^{-(t/RC)}$	$V_C = V_0(1 - e^{-(t/RC)})$ · $Q = Q_0(1 - e^{-(t/RC)})$ $V_R = V_0e^{-(t/RC)}$ · $I = I_0e^{-(t/RC)}$, with $I_0 = V_0/R$

$\tau = RC$ seconds: the time to fall to $1/e \approx 37\%$ of the initial value (or rise to 63%). $t_{1/2} = RC \ln 2$ · time for fraction f to remain: $t = -RC \ln f$.

Log-linear (the A* technique): $\ln I = \ln I_0 - t/RC$, so the gradient of **$\ln I$ against t is $-1/RC \rightarrow C = -1/(\text{gradient} \times R)$** . Derive it in two lines.

Derivation asked: $V_R = V_0e^{-(t/RC)}$ and by **Kirchhoff's second law** $V_0 = V_C + V_R$, so $V_C = V_0 - V_0e^{-(t/RC)}$.

THREE ROUTES TO C FROM A DISCHARGE GRAPH

1. Time constant: find $0.37V_0$, read off the time — that is $RC \rightarrow C = \tau/R$.

2. Half-life: time for V to halve $\rightarrow C = t_{1/2}/(R \ln 2)$.

3. Log route: plot $\ln V$ against t, gradient = $-1/RC$. **Most accurate — the route the scheme prefers on "determine" questions.**

4. Single pair into $V = V_0e^{-(t/RC)}$ — acceptable but less reliable; say so if asked to evaluate.

Tolerance questions: find the value, compute the allowed range ($\pm 10\%$ of $470\ \mu F \rightarrow 423\text{--}517\ \mu F$), **state explicitly whether it lies inside and conclude.**

For "maximum energy", use the **maximum** capacitance before $W = \frac{1}{2}CV^2$.

"EXPLAIN THE VARIATION OF V AS THE CAPACITOR CHARGES" — THE 6-MARK CHAIN

1. On closing the switch there is a current · **2.** the charge on the capacitor increases · **3.** the p.d. across the capacitor increases and **the p.d. across the resistor decreases** · **4.** since $I = V_R/R$, **the current decreases** · **5.** so the **rate of charging decreases** (the gradient falls) · **6.** eventually $V_C = V_0$, the current is zero and there is no further change.

Sketching the current graph: exponential decay from a maximum at $t = 0$, and you must **label the maximum as $I_0 = V_0/R$** . Asked repeatedly.

Second identical resistor in **parallel** $\rightarrow R$ halves $\rightarrow RC$ halves $\rightarrow$ charges faster. **The final charge is unchanged** — it depends only on C and V_0 .

SHEET 6 — MAGNETIC FIELDS

$F = BIl \sin\theta$ · $F = Bqv \sin\theta$ · $r = p/(BQ)$ · $\Phi = BA$ · flux linkage $N\Phi$ (Wb-turns) · $\mathcal{E} = -d(N\Phi)/dt$. Tesla base units $\text{kg A}^{-1} \text{s}^{-2}$; Wb = T m².

THE ANGLE TRAP — THE SINGLE MOST COMMON MCQ ERROR

In $F = BIl \sin\theta$, **θ is the angle between the current and the magnetic field.**

Wire **perpendicular** to $B \rightarrow \sin 90^\circ = 1 \rightarrow$ **maximum** force. Wire **parallel** to $B \rightarrow$ **zero** force.

In a square coil with its plane parallel to B , **sides parallel to B feel no force**; perpendicular sides feel BIl .

If the diagram gives the angle to the **plane** or to the **vertical**, convert carefully — a field at 20° to the vertical with a horizontal cable means **$\theta = 70^\circ$** .

Fleming's left hand: First = Field, seCond = Current, thuMb = Motion. For a **negative** charge the conventional current is **opposite** to the velocity — reverse your second finger.

Current balance: $F = \Delta m(\text{kg}) \times g$, then $B = F/(Il)$. **Convert grams to kilograms** — the classic slip. *Reaction explanation:* by Newton's third law the magnets feel an equal and opposite force, which is what the balance reads.

Moment on a coil: $F = BIlN$ on each long side; **moment = $F \times \text{width}$** . With a tilt, use the **perpendicular distance = width $\times \cos\theta$** . It rotates because by FLHR the forces on opposite sides are **up and down** (currents opposite), producing a couple.

DERIVE $r = p/BQ$ — WRITE BOTH LINES

The magnetic force provides the centripetal force: **$Bqv = mv^2/r \rightarrow Br = mv/q = p/q \rightarrow r = p/(Bq)$**

Find B from E_k : **$p = \sqrt{(2mE_k)}$** $\rightarrow B = p/(Qr)$. Relativistic: **$E = pc$** , so $p = E/c$. Specific charge: $e/m = v/(Br)$.

Mass spectrometer: after a velocity selector all ions share v , so $r = mv/(BQ)$ depends only on **m/Q** .

Ratios: alpha vs proton at the **same speed** $\rightarrow 4\times$ mass, $2\times$ charge $\rightarrow r_\alpha = 2r_p$. At the **same energy**, work through $p = \sqrt{(2mE_k)}$ — don't guess.

Reading particle tracks: curvature + FLHR + field direction gives the **sign of the charge**; opposite curvature = opposite charges.

Neutral particles leave NO track because they do not ionise the medium. Evidence one was produced: a **gap** between tracks · visible tracks **not collinear** with the incoming particle, so momentum requires an invisible particle · a **V-shape appearing in empty space**.

Direction of travel: the particle **loses momentum**, so $r = p/BQ$ **decreases** — the track spirals **inwards**, travelling from larger to smaller radius.

SHEET 7 — ELECTROMAGNETIC INDUCTION (the topic where wording matters most)

THE UNIVERSAL CHAIN — MEMORISE VERBATIM. SIX STATEMENTS = SIX MARKS.

1. There is a **change in (magnetic) flux linkage** with the coil (or: the conductor **cuts lines of magnetic flux**)
2. An **e.m.f. is induced** (Faraday: proportional to the **rate of change of flux linkage**)
3. There is a **complete circuit / conducting path**, so there is an **induced current**
4. The induced current produces **its own magnetic field**
5. By **Lenz's law the induced effect opposes the change that caused it** — producing a force opposing the motion
6. Consequence in context (the magnet slows / the balance reading rises / energy is dissipated / a bigger applied force is needed)

Context	The closer you must add
Copper tube with a slit	The slit reduces the number of paths for the induced current, so current, opposing force and time of fall are all smaller — the magnet falls faster
Wire on a balance	An upward force acts on the wire; by Newton's third law a downward force acts on the magnets, so the reading increases
Eddy-current brake	The induced current dissipates energy as heat ($P = I^2R$), so kinetic energy decreases. A thicker plate has lower resistance → larger current → larger opposing force
Induction cooker	a.c. → alternating magnetic field → changing flux linkage with the pan → induced e.m.f. → eddy currents in the pan → heat. The egg cooks only on contact because the egg is not a conductor
Rod reaching constant velocity	The induced e.m.f. opposes the battery e.m.f. , reducing the current; when they are equal the current and force are zero, so acceleration is zero
Current not rising instantly	The growing current creates changing flux linkage, inducing a back e.m.f. which by Lenz's law opposes the increase

Calculations: straight conductor $\epsilon = Blv$ · aeroplane wingtips use only the **vertical component** of the Earth's field (at the **equator the field is horizontal, so $\epsilon = 0$**) · coil in a changing field $\epsilon = NA(dB/dt)$ — find the **maximum gradient** of the B-t graph · rotating coil $\epsilon_{\max} = NBA\omega = \Phi\omega$ · **area under an ϵ -t graph = change in flux linkage** (because $\epsilon = d(N\Phi)/dt$). Remember $A = \pi r^2 = \pi(d/2)^2$.

THE A.C. GENERATOR GRAPH

e.m.f. is **maximum when the plane of the coil is PARALLEL to the field** — the sides cut flux at the maximum rate; flux linkage is momentarily zero but **changing fastest**.

e.m.f. is **zero when the plane is PERPENDICULAR** — the sides move parallel to the field lines and cut no flux; flux linkage is maximum but **momentarily unchanging**.

Doubling the speed: frequency doubles and peak e.m.f. doubles — **both must be shown**.

Transformer chain: alternating p.d. → alternating current in the primary → alternating magnetic field in the core → change of flux linkage with the secondary → induced e.m.f. → complete circuit → current. **Why the secondary p.d. is less than expected: not all the flux produced by the first coil links with the second** (flux leakage).

SHEET 8 — ALPHA SCATTERING, de BROGLIE & ACCELERATORS

Observation	Conclusion
The vast majority pass straight through undeflected	The atom is mostly empty space
Some are deflected through small angles	There is a concentration of charge in the atom
A very small number are deflected through more than 90°	Almost all the mass is concentrated in a very small volume — the nucleus — and the charge concentration is large

THE MCQ TRAP YOU MUST NEVER FALL FOR

"**The nucleus is made up of protons and neutrons**" is **NOT a conclusion** — the neutron was not discovered until 1932, long after 1911. This distractor appears in at least six papers.
 Also invalid: "the nucleus is **positively** charged" — the scattering **cannot determine the sign**, because a negative central charge would **attract** the alpha and a hyperbolic attractive orbit can also deflect it through a large angle. Also invalid: "electrons orbit in energy levels".

Why the foil must be very thin: so alphas pass through **few layers of atoms** and are unlikely to undergo **multiple deflections** or be absorbed — alphas are strongly ionising with very low penetration. **Why a vacuum:** alphas would be absorbed by air.

Why alpha, not beta or gamma: beta is far **less massive** so would be deflected by atomic electrons too, and is more penetrating; **gamma is uncharged** so experiences no electrostatic force; alpha has **double the charge** so experiences a larger force.

de Broglie: $\lambda = h/p$, and with $eV = \frac{1}{2}mv^2$, $\lambda = h/\sqrt{2meV}$. Rings get **smaller** when λ gets smaller, i.e. when momentum/energy **increases**.

Why high energies probe nucleons (3 marks): high energy $\rightarrow$ **large momentum** $\rightarrow \lambda = h/p$ so a **very small de Broglie wavelength, comparable to a nucleon's diameter** $\rightarrow$ **better resolution**. **MCQ traps:** "to produce individual quarks" is wrong — quark confinement. "To ionise nucleons" is wrong.

LINAC — 6-MARK * ANSWER

1. Charged particles are accelerated by the **electric field in the gaps between the drift tubes** · **2.** there is **no electric field inside a drift tube**, so the particle travels at constant speed while inside (it is shielded) · **3.** the a.c. supply has a **constant frequency** · **4.** the **polarity reverses while the particle is inside a tube**, so the field is always accelerating at the next gap · **5.** the particle must spend the **same time in each tube** (half a period); since its speed increases, $L = vT/2$ means the **tubes get progressively longer** · **6.** at the end the particles travel close to **c** and their speed no longer increases appreciably, so the tubes are of **constant length**.

CYCLOTRON — 6-MARK * ANSWER

1. The alternating p.d. between the dees creates an **electric field in the gap** · **2.** which **accelerates the particles as they cross the gap**, so they gain kinetic energy · **3.** inside the dees there is **no electric field**; the magnetic field exerts a force **perpendicular to the velocity** · **4.** this provides the **centripetal force**, so the path is circular · **5.** the **polarity reverses every half cycle** while the particle is inside a dee · **6.** so the field always accelerates it across the gap; **$r = mv/Bq$ increases each time**, producing a spiral until it leaves at the maximum radius.

$f = Bq/(2\pi m)$ · **$T = 2\pi m/(Bq)$** · **time in ONE dee = $\pi m/(Bq)$** · exit: $p = BQr_{\max}$, $E_k = (BQr)^2/(2m)$.

The frequency is independent of speed and radius — this is why a constant-frequency supply works. A favourite MCQ.

Colliding beams vs a fixed target (4 marks): with two beams of **equal and opposite momentum the total momentum before the collision is zero** · momentum must be conserved, so the **products can be created at rest** — **all** the kinetic energy is available to be converted into mass · with a stationary target the system has net momentum, so the products must **carry kinetic energy away**, leaving less for mass · therefore colliding beams produce **larger masses**.

SHEET 9 — THE STANDARD MODEL & CONSERVATION LAWS

Quarks u, c, t (+2/3) · d, s, b (−1/3) — fundamental. **Leptons** e^-, μ^-, τ^- (−1), ν_e, ν_μ, ν_τ (0) — fundamental.

Hadrons are made of quarks and are **NOT fundamental**: **baryons** = **3 quarks** ($p = uud, n = udd, \Lambda^0 = uds, \Sigma^+ = uus, \Omega^- = sss$); **mesons** = **1 quark + 1 antiquark** ($\pi^+ = u\bar{d}, \pi^- = d\bar{u}, K^- = s\bar{u}$).

Say in one line each: "A baryon is made of **three quarks**." · "A meson is made of **a quark and an antiquark**." · "**Leptons are fundamental** — they are not made of quarks."

Antiparticles: same mass, same magnitude of charge but **opposite sign**, and opposite baryon/lepton number. **Trap**: an electron and a positron do **not** differ in baryon number — both have $B = 0$. Antiproton mass is **positive** (mass is never negative).

Law	Rule
Charge	Total before = total after
Baryon number	Baryons +1, antibaryons −1, mesons and leptons 0
Lepton number	Checked separately for each generation — L_e, L_μ, L_τ . Particles +1, antiparticles −1
Energy / momentum	Including rest mass-energy; momentum is a vector

THE A* DISCRIMINATOR — LEPTON NUMBER PER FAMILY

$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$ is **allowed** ($L_e: 0 \rightarrow 1 - 1 = 0 \checkmark$; $L_\mu: 1 \rightarrow 1 \checkmark$).

$\mu^- \rightarrow e^- + \nu_e + \bar{\nu}_\mu$ is **forbidden** ($L_e: 0 \rightarrow +2 \ X$; $L_\mu: 1 \rightarrow -1 \ X$).

Lay the answer out line by line — marks are for the numbers on each side, not just the verdict — then conclude: "Both charge and lepton number are conserved, **so the decay is possible**."

Decays to know by heart — the bar placement is asked almost every session:

β^- : $n \rightarrow p + e^- + \bar{\nu}_e$ (an **ANTI**-neutrino) · β^+ : $p \rightarrow n + e^+ + \nu_e$ (a neutrino) · $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$ · $\pi^+ \rightarrow \mu^+ + \nu_\mu$ · $\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$.

Annihilation: particle + antiparticle → **TWO photons, so that momentum is conserved**. From rest, energy per photon = mc^2 (electron: 8.2×10^{-14} J = 0.511 MeV); if moving, add the kinetic energies **before** dividing by 2.

Pair production: minimum photon energy = $2mc^2 = 1.02$ MeV for e^+e^- . **Do the comparison explicitly**.

Track evidence for pair production (6-mark *): the photon leaves **no track because it has no charge** · tracks are produced by charged particles · the two curve in **opposite directions** → the charges are opposite · the **radii are the same** → since $r = p/BQ$, equal radii and equal charge magnitude mean **equal momenta**, so equal masses — consistent with a particle-antiparticle pair.

Mass-energy: $\Delta E = c^2 \Delta m$. MeV/ c^2 → kg: ($E \times 1.6 \times 10^{-13}$)/ c^2 . Rest energies worth memorising: **electron 0.511 MeV · proton 938 MeV ≈ 0.94 GeV · muon 106 MeV · charged pion ~140 MeV**.

Why GeV/c is a unit of momentum: because $E = pc$, so $p = E/c$ — energy divided by speed has the units of momentum.

Maximum mass created: $m_{\max} = E_{\text{total}}/c^2$, using **all** kinetic plus rest energy.

RELATIVITY — YOU ARE NOT ASKED TO CALCULATE γ

The required method: **1.** do the **classical** calculation (distance = speed × lifetime) · **2.** compare with the observation · **3.** explain the discrepancy **in words**.

Time dilation: the lifetime measured in the **laboratory frame is longer** than the proper lifetime, so the rate of decay decreases and more survive.

Length contraction (particle's frame): the distance to the ground is **contracted**, so it is covered within the normal lifetime. **Be consistent about which frame you are in**.

Mass increase: as $v \rightarrow c$ the mass increases, so the speed can approach but **never reach c** — why linac drift tubes stop lengthening.

The muon answer: speed $\approx c$ · classically it would travel only **656 m**, far less than 1600 m · so classically almost none should arrive, yet 74% do · because of relativistic speed the **lifetime is dilated in the Earth's frame** (or the distance is contracted in theirs) · so most survive — the observations are **consistent with relativity**.

SHEET 10 — WHAT IS NOT ON THE FORMULA SHEET

YOU MUST KNOW THESE — THEY ARE NOT PRINTED

$\omega = 2\pi f$ and $\omega = 2\pi/T$ (only $T = 2\pi/\omega$ is printed)

$W = \frac{1}{2}CV^2$ and $W = \frac{1}{2}Q^2/C$ (only $W = \frac{1}{2}QV$ is printed)

$V = V_0 e^{-(t/RC)}$ and $I = I_0 e^{-(t/RC)}$ (only $Q = Q_0 e^{-(t/RC)}$ is printed)

$V_C = V_0(1 - e^{-(t/RC)})$ and $V_R = V_0 e^{-(t/RC)}$

$\tau = RC$ and $t_{1/2} = RC \ln 2$

Capacitors in series and parallel

$C = 4\pi\epsilon_0 r$ for an isolated sphere

$V = kQ/r$ — **the potential of a point charge. The field equation is printed, the potential one is not. This is the most commonly forgotten formula on the paper.**

$W = QV \cdot \phi = BA$ and flux linkage = $NBA \cdot \epsilon = Blv \cdot \epsilon_{\max} = NBA\omega$

$f = Bq/(2\pi m)$ (cyclotron) $\cdot E = pc \cdot$ quark charges and the standard model classification $\cdot$ base-unit derivations

Quantity	Base units
Momentum / impulse	kg m s^{-1} (= N s)
Capacitance (F)	$\text{kg}^{-1} \text{m}^{-2} \text{A}^2 \text{s}^4$
Magnetic flux density (T)	$\text{kg A}^{-1} \text{s}^{-2}$
Magnetic flux (Wb)	$\text{T m}^2 = \text{kg m}^2 \text{A}^{-1} \text{s}^{-2}$
Planck constant (J s)	$\text{kg m}^2 \text{s}^{-1}$

Scalars: capacitance, **electric potential**, energy, mass, speed, charge, time, density. **Vectors:** momentum, impulse, **electric field strength**, magnetic flux density, centripetal acceleration, force, velocity. **Trap:** **potential is scalar, field strength is vector.**

SHEET 11 — COMMAND WORDS & PHRASE BANK

Command	What the mark scheme wants
State	A single fact or name. No explanation — do not waste time
Show that...	Full working to a value; quote your value and compare with the target . Then use your unrounded value , not the rounded target, later
Determine	Calculate, usually extracting data from a graph or diagram first
Deduce	Calculate and state a reasoned conclusion in a sentence — the conclusion carries its own mark
Explain	Physics reasons , linked with because/so/therefore. A description without causation scores zero
Discuss / Assess / Evaluate	Both sides, applied physics, and an explicit judgement — say whether the claim is correct
Derive	Start from a stated principle and show every algebraic step
Compare	Comparative language for both items

PHRASE BANK — LEARN VERBATIM

Momentum: "Total momentum remains constant **provided no resultant external force acts.**" $\cdot$ "Elastic means **kinetic energy is conserved.**" $\cdot$ "Friction is negligible **so there is no resultant external force.**"

Circular: "The **resultant force acts towards the centre.**" $\cdot$ "A force is required to **change the direction** of motion." $\cdot$ "There is no outward force on the passenger." $\cdot$ "The magnitude of the centripetal force is constant **because the speed is constant.**"

Electric fields: "Electric field strength is the **force per unit positive charge.**" $\cdot$ "All the kinetic energy is converted to electric potential energy at the closest approach." $\cdot$ "Assume the nucleus behaves as a **point charge.**"

Capacitors: "The p.d. across the capacitor increases, **so the p.d. across the resistor decreases, so the current decreases.**" $\cdot$ "The time constant is the time for the p.d. to fall to **1/e (37%)** of its initial value."

Induction: "There is a **change in magnetic flux linkage** with the coil." $\cdot$ "The conductor **cuts lines of magnetic flux.**" $\cdot$ "An **e.m.f. is induced.**" $\cdot$ "There is a **complete circuit**, so there is a current." $\cdot$ "By **Lenz's law the induced e.m.f. opposes the change that caused it.**" $\cdot$ "Not all the flux produced by the first coil links with the second coil."

Particles: "A baryon is **three quarks**; a meson is **a quark and an antiquark.**" $\cdot$ "Leptons are fundamental." $\cdot$ "Neutral particles do not ionise, so they leave no track." $\cdot$ "High momentum gives a **short de Broglie wavelength, which gives better resolution.**" $\cdot$ "The lifetime is dilated in the laboratory frame because the speed is close to c."

Alpha scattering: "Most pass straight through undeflected, **so the atom is mostly empty space.**" $\cdot$ "A few are deflected through more than 90°, so there is a concentration of charge and mass in a very small volume."

SHEET 12 — THE CHECKLIST & EXAM-DAY TACTICS

Tick each only when you can do it **cold, in under the mark-allocation time**.

Can you...	Can you...
- Resolve a 2D collision into components and find speed and angle? - Deduce elastic/inelastic with an explicit numerical comparison? - Derive $\mathbf{a} = \mathbf{v}^2/r$ with the vector diagram — all five steps? - Do the conical pendulum / banked track with the full free-body treatment? - Do a vertical circle: max and min contact force, minimum speed? - Find the closest approach of an alpha, then the force and acceleration? - Draw radial and uniform fields with equipotentials? - Combine field (vector) and potential (scalar) from two point charges? - Find C from a discharge graph by three routes, including log-linear? - Sketch charging/discharging graphs with $I_0 = V_0/R$ labelled?	- Write the 6-mark capacitor-charging chain? - Write the full induction chain plus the context closer? - Get ϵ from a B-t or ϕ-t graph using the maximum gradient? - Explain a linac and a cyclotron as 6-mark answers, plus $f = Bq/2\pi m$? - Check charge, baryon and lepton number line by line and conclude? - Give alpha scattering as three matched observation-conclusion pairs? - Convert $\text{MeV}/c^2 \leftrightarrow \text{kg}$ and compare with a stated claim? - Do the muon calculation and the relativistic explanation? - Read particle tracks — sign, direction, neutral particles, momentum? - Give base units of impulse, farad, tesla and weber?

EXAM-DAY TACTICS

- Section A first, fast** — aim for 10 minutes. Every distractor exists to test **one specific misconception**; most are in these sheets.
- Read the mark allocation as an instruction.** A 2-mark "explain" wants exactly two distinct physics statements; a 6-mark * wants six.
- Write on the graph.** Tangents, triangles, read-off lines, the value of $0.37V_0$ — those marks are awarded **from the graph**, not the text.
- Never leave a "show that" unconcluded**, and never use the rounded show-that value if you calculated a better one.
- On * questions use numbered points with connectives** — two of the six marks are for **structure and sustained reasoning**; a disconnected bullet list caps you at 4.
- State the conclusion sentence on every deduce/discuss/assess** — the single most commonly dropped mark on the paper.
- If the question asks for magnitude AND direction, there is a separate mark for direction.**
- Carry unrounded values** through multi-step calculations.
- If stuck on an early part, invent a sensible value and continue** — ecf means later parts are still fully available.
- Leave 5 minutes** to check units, significant figures, and that every "deduce" has a conclusion.